Varied prevalence of Clostridium difficile in an integrated swine operation.

The objectives of this study were to compare the prevalence of Clostridium difficile (Cd) among different age and production groups of swine in a vertically integrated swine operation in Texas in 2006 and to compare our isolates to other animal and human isolates. Results are based on 131 Cd isolates from 1008 swine fecal samples and pork trim samples (overall prevalence of 13%). The prevalence (number positive/number tested in production type) of Cd was different between the groups (P<or=0.001), and was highest among suckling piglets at 50.0% (61/122), followed by 23.8% (34/143) for lactating sows and effluent from the farrowing barn, 8.4% (10/119) for nursery, 6.5% (4/62) for pork products, 3.9% (15/382) for grower-finisher, and 3.9% (7/180) for breeding boars and sows. Of the 131 isolates, 122 were positive by PCR for both toxins A (tcdA) and B (tcdB) genes, 129 isolates harbored a 39 base pair deletion in the tcdC gene, 120 isolates were toxinotype V, and all 131 of the isolates were positive for the binary toxin gene cdtB. All isolates were resistant to cefoxitin, ciprofloxacin, and imipenem, whereas all were sensitive to metronidazole, piperacillin/tazobactam, amoxicillin/clavulanic acid, and vancomycin. The majority of isolates were resistant to clindamycin; resistant or intermediate to ampicillin; and sensitive to tetracycline and chloramphenicol. There was an increased (P</=0.001) number of isolates for the timeframe of September to February compared to March to August.